

SPACE / VACUUM (T = 100 K)

Surface $T_s = 100$ K

Cycloidal Ridges & Fractures

Upwarped Dome

Disrupted Chaos / Pit

STAGNANT LID (RIGID ICE)
 $\delta_{lid} \approx 10 - 15$ km (~ 60% of total shell)
Heat Transport: Pure Conduction ($F_{cond} = k \frac{\Delta T}{D}$)
Viscosity: $\eta(T) > 10^{16}$ Pa s ($T = 100$ K $\rightarrow$ 240 K)

Rheological Boundary $T_{lid} = 240$ K

Thermal Diapir
 $\sigma_{buoy} \approx 5$ kPa

Thermal Diapir
 $\tau \sim 10^4$ yr

Cold Downwelling

**CONVECTIVE
SUBLAYER**
 $\delta_{conv} \approx 5 - 10$ km
 $\eta_b \approx 10^{14}$ Pa s
 $u \approx 0.5$ m/yr

Ice-Ocean Melting Boundary $T_{base} = 270$ K

SUBSURFACE LIQUID WATER OCEAN
Global Decoupled Ocean ($T_{base} \approx 270$ K, Depth ~ 100 km)

Total Shell Thickness $D \approx 20$ km